

A ROOT MUSIC parameter estimation algorithm based on tensor decomposition for FDA-MIMO radar

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Abstract—In parameter estimation algorithms for frequency diverse array and multiple-input multiple-output (FDA-MIMO) radar, traditional subspace-based algorithms using matrix decomposition suffer from low estimation accuracy and cannot meet the high-precision requirements of practical applications. Therefore, this paper proposes a tensor decomposition-based root multiple signal classification (root-MUSIC) parameter estimation algorithm for FDA-MIMO radar. The proposed algorithm first stacks the received data of the FDA-MIMO radar into a third-order tensor. Then, the signal subspace is obtained using higher-order singular value decomposition (HOSVD), and a two-dimensional (2D) spatial spectrum function is constructed. Subsequently, the root-MUSIC technique is employed to avoid 2D spectral peak search (SPS) and obtain the estimates of angle and range. Experimental results validate the superiority and effectiveness of the proposed method.

Index Terms—Root MUSIC, FDA-MIMO radar, Tensor decomposition, Parameter estimation.

I. INTRODUCTION

Multiple-input multiple-output (MIMO) radar transmits mutually orthogonal signals through multiple transmit antennas and captures echo signals exploiting multiple receive antennas, followed by matched filtering, which significantly increases the degrees of freedom (DOFs) [1]–[3]. However, conventional

MIMO radar systems employ the same carrier frequency at the transmitter, which prevents them from distinguishing targets at the same angle but varying ranges. In contrast, frequency diverse array (FDA) introduces small frequency offsets between adjacent antennas, generating range-angle dependent beampattern that enable the capability to distinguish targets at the same angle but varying ranges. To endow MIMO radar with range resolution capability, researchers have configured the transmitter as an FDA, thereby designing the FDA-MIMO radar system [4]. Due to its capability for joint range and angle estimation, parameter estimation for FDA-MIMO is currently a major research focus.

Subspace-based algorithms were among the earliest approaches employed for parameter estimation in FDA-MIMO radar systems. In [5], a two-dimensional multiple signal classification (2D-MUSIC) algorithm was introduced for parameter estimation in monostatic FDA-MIMO radar. However, this algorithm suffers from both high computational complexity and off-grid errors due to the required 2D spectral peak search (SPS). To improve computational efficiency, a reduced-dimension (RD)-MUSIC technique for joint range and angle parameter estimation was developed [6]. While effective, this RD-MUSIC algorithm still exhibits off-grid errors. Therefore, in [7], an RD-root-MUSIC method was applied to monostatic FDA-MIMO radar parameter estimation to derive closed-form solutions for range and angle estimates. Compared to MUSIC techniques, estimation of signal parameters via rotational invariance technique (ESPRIT) techniques offer lower compu-

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tational complexity [8]. Accordingly, to reduce computational complexity, an approach utilizing two ESPRIT operations was proposed in [9] to obtain range and angle estimates for FDA-MIMO radar. Furthermore, unitary ESPRIT algorithms have been developed for both monostatic [10] and bistatic FDA-MIMO radar systems [11], which further reduce computational complexity by leveraging real-valued computations. However, all the aforementioned subspace-based algorithms rely on matrix-based approaches and are incapable of exploiting the inherent multidimensional data structure. In contrast, tensor decomposition is a technique specifically developed for multidimensional data, which can leverage the multidimensional structural information to achieve performance improvements. Higher-order singular value decomposition (HOSVD) is a widely used tensor decomposition technique that has demonstrated excellent performance in many fields. Therefore, to further enhance parameter estimation accuracy in FDA-MIMO radar, a RD-MUSIC algorithm based on HOSVD was proposed in [12]. Unfortunately, this RD-MUSIC algorithm based on HOSVD still suffers from off-grid errors.

This paper presents a tensor root-MUSIC algorithm designed for the joint estimation of range and angle. The proposed framework begins by modeling the received data as a third-order tensor and applying HOSVD to obtain the signal subspace. Then, the RD-MUSIC technique is employed to avoid the 2D-SPS, yielding a one-dimensional (1D) spatial spectrum function. Subsequently, angle estimates are obtained through polynomial rooting. Finally, the transmit steering vector is derived through partial derivatives of the Lagrange function, and range estimates are obtained using the least squares fitting technique. Simulation results confirm that the proposed method yields superior estimation accuracy.

Notations: The symbols $\otimes$, $\oplus$, $\odot$, and $\circ$ denote the Kronecker, Hadamard, Khatri-Rao, and outer products, respectively. Lowercase letter a , lowercase bold italic letter $\mathbf{a}$, uppercase bold italic letter $\mathbf{A}$, and Calligraphic bold italic letter $\mathcal{A}$ represent scalars, vectors, matrices, and tensors, respectively. The inverse, conjugation, pseudoinverse, transpose, and Hermitian transpose are represented by $(\cdot)^{-1}$, $(\cdot)^*$, $(\cdot)^\dagger$, $(\cdot)^T$, and $(\cdot)^H$, respectively. $\mathbf{I}_N$ is the $N \times N$ identity matrix. $\angle(\cdot)$, $\text{normal}(\cdot)$, $\det(\cdot)$, and $\text{comp}(\cdot)$ denote the complex phase angle, vector normalization, matrix determinant, and companion matrix, respectively. $[\mathcal{A}]_v$ stands for the mode- v unfolding of $\mathcal{A}$. $\|\cdot\|_F$ and $\min(\cdot)$ are the Frobenius norm and minimum value symbols, respectively. $\mathcal{A} \times_v \mathbf{B}$ denotes the v -mode product of $\mathcal{A}$ and $\mathbf{B}$.

II. TENSOR SIGNAL MODEL

Consider a monostatic FDA-MIMO radar system illustrated in Fig. 1. Both the transmitter and receiver consist of uniform linear arrays (ULAs) containing M and N antennas, respectively, with an inter-element spacing of d . With f_0 serving as the base carrier frequency of the first transmit element, the transmit frequency for the m -th antenna is:

$$f_m = f_0 + (m-1)\Delta f \quad (1)$$

where $\Delta f \ll f_0$ is the frequency offset.

The transmit signal of the m -th antenna is formulated as:

$$\bar{h}_m(t) = \psi_m(t)e^{j2\pi f_m t}, \quad 0 < t < T \quad (2)$$

where T and $\psi_m(t)$ are the pulse duration and transmitting waveform, respectively. Considering the mutual orthogonality of the transmitted waveforms, it yields:

$$\int_0^T \psi_m(t)\psi_{m'}^*(t-\tau)e^{j2\pi(m-m')\Delta f t} dt = \begin{cases} 1, & m = m', \forall \tau \\ 0, & m \neq m', \forall \tau \end{cases} \quad (3)$$

Consider a scenario with P uncorrelated far-field targets. For

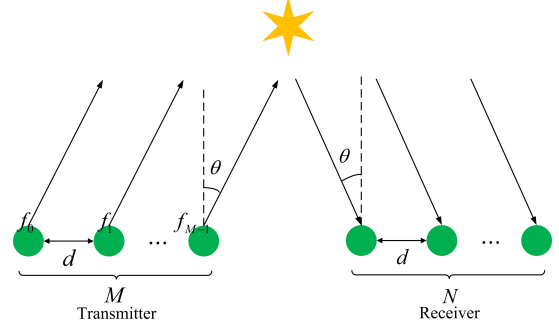


Fig. 1. The monostatic FDA-MIMO radar

the p -th target ($p = 1, \dots, P$), r_p and θ_p represent the range and angle, respectively. Upon applying matched filtering, the resulting signal is [5]:

$$\mathbf{x}(q) = (\mathbf{B}_R \odot \mathbf{B}_T)\mathbf{s}(q) + \mathbf{n}(q) \in \mathbb{C}^{MN \times 1} \quad (4)$$

where $\mathbf{s}(q) = [s_1(q), \dots, s_P(q)]^T \in \mathbb{C}^{P \times 1}$ with $s_p(q) = \sigma e^{j2\pi f_d(q-\tau_0)} e^{-j4\pi f_0 r_p/c}$ and $\mathbf{n}(q)$ is the noise vector. It is worth noting that $\tau_0 = 2r/c$ with c being the speed of light, f_d denotes the corresponding normalized Doppler frequency, and σ denotes the radar cross section. In addition, $\mathbf{B}_R = [\mathbf{b}_r(\theta_1), \dots, \mathbf{b}_r(\theta_P)] \in \mathbb{C}^{N \times P}$ and $\mathbf{B}_T = [\mathbf{b}_t(r_1, \theta_1), \dots, \mathbf{b}_t(r_P, \theta_P)] \in \mathbb{C}^{M \times P}$ are the receive and transmit steering matrices, respectively. Since $d = c/f_N = \lambda_0/2$, the transmitting steering vector $\mathbf{b}_t(r_p, \theta_p)$ and the receiving steering vector $\mathbf{b}_r(\theta_p)$ are formulated as:

$$\mathbf{b}_t(r_p, \theta_p) = [1, e^{j\pi(\sin\theta_p - 4\Delta f r_p/c)}, \dots, e^{j\pi(M-1)(\sin\theta_p - 4\Delta f r_p/c)}]^T \in \mathbb{C}^{M \times 1} \quad (5a)$$

$$\mathbf{b}_r(\theta_p) = [1, e^{j\pi\sin\theta_p}, \dots, e^{j\pi(N-1)\sin\theta_p}]^T \in \mathbb{C}^{N \times 1} \quad (5b)$$

Collecting Q snapshots yields the received signal matrix:

$$\mathbf{X} = [\mathbf{x}(1), \dots, \mathbf{x}(Q)] = (\mathbf{B}_R \odot \mathbf{B}_T)\mathbf{S} + \mathbf{N} \in \mathbb{C}^{MN \times Q} \quad (6)$$

where $\mathbf{N} = [\mathbf{n}(1), \dots, \mathbf{n}(Q)] \in \mathbb{C}^{MN \times Q}$ is the noise matrix and $\mathbf{S} = [\mathbf{s}(1), \dots, \mathbf{s}(Q)] \in \mathbb{C}^{MN \times Q}$ is the signal matrix.

By exploiting the trilinear model, $\mathbf{X} \in \mathbb{C}^{M \times N \times Q}$ in (6) is reformulated as a third-order tensor [14]:

$$\mathcal{X} = \sum_{p=1}^P \mathbf{b}_t(\theta_p, r_p) \circ \mathbf{b}_r(\theta_p) \circ \bar{\mathbf{s}}(p) + \mathcal{N} \quad (7)$$

where $\mathcal{N} \in \mathbb{C}^{M \times N \times Q}$ is the noise tensor.

III. ROOT-MUSIC PARAMETER ESTIMATION TECHNIQUE BASED ON TENSOR DECOMPOSITION

A. Signal Subspace Estimation with HOSVD

According to [14], the HOSVD of $\mathcal{X}$ is:

$$\mathcal{X} = \mathcal{G} \times_1 \mathbf{W}_1 \times_2 \mathbf{W}_2 \times_3 \mathbf{W}_3 \quad (8)$$

where $\mathcal{G}$ represents the core tensor obtained from HOSVD. $\mathbf{W}_v$ denotes the matrix of left singular vectors corresponding to the mode- v unfolding of $\mathcal{X}$, where $v \in \{1, 2, 3\}$. Since the rank of $\mathcal{X}$ equals the number of targets, the tensor-based subspace can be obtained by the truncating HOSVD technique, specifically [15]:

$$\mathcal{X}_T = \mathcal{G}_T \times_1 \mathbf{W}_{t1} \times_2 \mathbf{W}_{t2} \quad (9)$$

where $\mathcal{G}_T = \mathcal{X} \times_1 \mathbf{W}_{t1}^H \times_2 \mathbf{W}_{t2}^H \times_3 \mathbf{W}_{t3}^H$ is the truncated core tensor. Moreover, $\mathbf{W}_{tv}$ ($v = 1, 2, 3$) is obtained by selecting the first P principal singular vectors from $\mathbf{W}_v$ ($v = 1, 2, 3$). Following this, we can obtain:

$$\mathcal{X}_T = \mathcal{X} \times_1 (\mathbf{W}_{t1} \mathbf{W}_{t1}^H) \times_2 (\mathbf{W}_{t2} \mathbf{W}_{t2}^H) \times_3 \mathbf{W}_{t3} \quad (10)$$

The signal subspace can be constructed by applying mode-3 tensor unfolding to $\mathcal{X}_T$, as follows [13]:

$$\mathbf{E}_S = [\mathcal{X}_T]_3^T = (\mathbf{W}_{t2} \mathbf{W}_{t2}^H \otimes \mathbf{W}_{t1} \mathbf{W}_{t1}^H) [\mathcal{X}]_3^T \mathbf{W}_{t3}^* \quad (11)$$

Due to $[\mathcal{X}]_3^T \approx \mathbf{V}_{t3}^* \mathbf{\Lambda}_{t3} \mathbf{V}_{t3}^T$, we have:

$$\mathbf{E}_S = (\mathbf{W}_{t2} \mathbf{W}_{t2}^H \otimes \mathbf{W}_{t1} \mathbf{W}_{t1}^H) \mathbf{\Lambda}_{t3}^* \mathbf{V}_{t3} \quad (12)$$

B. Angle estimation based on Root-MUSIC

The noise subspace is derived by orthogonalizing the signal subspace. The specific operations are as follows:

$$\mathbf{E}_{orth} = \mathbf{E}_N \mathbf{E}_N^H = \mathbf{I}_{MN} - \mathbf{E}_0 \mathbf{E}_0^H \quad (13)$$

where $\mathbf{E}_0$ denotes the orthogonal basis of $\mathbf{E}_S$.

Then, the 2D spatial spectrum function is formulated as:

$$f(r, \theta) = \frac{1}{[\mathbf{b}_r(\theta) \otimes \mathbf{b}_t(r, \theta)]^H \mathbf{E}_{orth} [\mathbf{b}_r(\theta) \otimes \mathbf{b}_t(r, \theta)]} \quad (14)$$

By performing a 2D-SPS on (14), estimates of angle and range can be obtained. However, the computational complexity of 2D-SPS is prohibitively high, rendering it inefficient for practical applications. Therefore, this paper will subsequently introduce an RD-root-MUSIC technique to avoid the high computational burden of the 2D peak search procedure.

Define

$$\begin{aligned} \mathbf{Z}(r, \theta) &= [\mathbf{b}_r(\theta) \otimes \mathbf{b}_t(r, \theta)]^H \mathbf{E}_{orth} [\mathbf{b}_r(\theta) \otimes \mathbf{b}_t(r, \theta)] \\ &= \mathbf{b}_t(r, \theta)^H [\mathbf{b}_r(\theta) \otimes \mathbf{I}_N]^H \mathbf{E}_{orth} [\mathbf{b}_r(\theta) \otimes \mathbf{I}_N] \mathbf{b}_t(r, \theta) \\ &= \mathbf{b}_t(r, \theta)^H \mathbf{D}(\theta) \mathbf{b}_t(r, \theta) \end{aligned} \quad (15)$$

where $\mathbf{D}(\theta) = [\mathbf{b}_r(\theta) \otimes \mathbf{I}_N]^H \mathbf{E}_{orth} [\mathbf{b}_r(\theta) \otimes \mathbf{I}_N]$. By introducing a constraint condition $e^H \mathbf{b}_t(r, \theta) = 1$, the optimization problem of (14) can be expressed as:

$$\min_{r, \theta} \mathbf{b}_t(r, \theta)^H \mathbf{D}(\theta) \mathbf{b}_t(r, \theta) \quad \text{s.t.} \quad e^H \mathbf{b}_t(r, \theta) = 1 \quad (16)$$

where $e = [1, 0, \dots, 0]^T \in \mathbb{R}^{M \times 1}$ is a vector whose first element is 1 and the rest are 0, used to eliminate the solution $\mathbf{b}_t(r, \theta) = \mathbf{0}_M$. By applying the Lagrangian multiplier, we can derive:

$$L(r, \theta) = \mathbf{b}_t(r, \theta)^H \mathbf{D}(\theta) \mathbf{b}_t(r, \theta) - \mu (e^H \mathbf{b}_t(r, \theta) - 1) \quad (17)$$

where μ is a constant. Computing the partial derivative of (17) yields:

$$\partial L(r, \theta) / \partial \mathbf{b}_t(r, \theta) = 2 \mathbf{D}(\theta) \mathbf{b}_t(r, \theta) - \mu e = 0 \quad (18)$$

Combining (18) with $e^H \mathbf{b}_t(r, \theta) = 1$ leads to $\mu/2 = 1/e^H \mathbf{D}(\theta)^{-1} e$. Hence, we get:

$$\mathbf{b}_t(r, \theta) = \frac{\mathbf{D}(\theta)^{-1} e}{e^H \mathbf{D}(\theta)^{-1} e} \quad (19)$$

Subsequently, the optimization problem in (16) is equivalent to:

$$\min_{\theta} \frac{1}{e^H \mathbf{D}(\theta)^{-1} e} = \min_{\theta} \frac{\det(\mathbf{D}(\theta))}{\text{comp}(\mathbf{D}(\theta))_{1,1}} \quad (20)$$

where $\text{comp}(\cdot)$ represents the companion matrix computation. Let $z = e^{j\pi \sin \theta_p}$, reformulating (20) into a polynomial rooting problem, we obtain:

$$\det(\mathbf{D}(z)) = 0 \quad (21)$$

where $\mathbf{D}(z) = [\mathbf{b}_r(z) \otimes \mathbf{I}_N]^H \mathbf{E}_{orth} [\mathbf{b}_r(z) \otimes \mathbf{I}_N]$. By replacing $\mathbf{b}_r^T(z^{-1})$ with $\mathbf{b}_r^H$, $\mathbf{D}(z)$ can be rewritten as:

$$\mathbf{D}(z) = [z^{M-1} \mathbf{b}_r(z^{-1}) \otimes \mathbf{I}_N]^T \mathbf{E}_{orth} [\mathbf{b}_r(z) \otimes \mathbf{I}_N] \quad (22)$$

Finally, the angle estimate is given by:

$$\hat{\theta}_p = \arcsin[\text{angle}(\hat{z}_p)/\pi] \quad (23)$$

where $\hat{z}_p$ is the root of (21).

C. Range estimation based on least squares fitting

Based on (19), the estimate of the transmit steering vector is given by:

$$\hat{\mathbf{b}}_t(r_p, \hat{\theta}_p) = \frac{\mathbf{D}(\hat{\theta}_p)^{-1} e}{e^H \mathbf{D}(\hat{\theta}_p)^{-1} e} \quad (24)$$

Utilizing the obtained angle estimate to eliminate the angle component in $\hat{\mathbf{b}}_t(r_p, \hat{\theta}_p)$ yields the range component of $\hat{\mathbf{b}}_t(r_p, \hat{\theta}_p)$:

$$\begin{aligned} \hat{\mathbf{b}}_t(r_p) &= \hat{\mathbf{b}}_t(r_p, \hat{\theta}_p) \oplus \hat{\mathbf{b}}_t'(\hat{\theta}_p) \\ &= [1, e^{-j4\pi \Delta f r_p/c}, \dots, e^{-j4(M-1)\pi \Delta f r_p/c}]^T \end{aligned} \quad (25)$$

where $\hat{\mathbf{b}}_t'(\hat{\theta}_p) = [1, e^{-j\pi \sin \theta_p}, \dots, e^{-j(M-1)\pi \sin \theta_p}]^T$.

Next, a matrix is constructed as follows:

$$\mathbf{O} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ 0 & \pi & \dots & (M-1)\pi \end{bmatrix} \quad (26)$$

Using the least squares fitting approach, we can derive:

$$\min_{\kappa_p} \|\mathbf{O}^T \kappa_p - \hat{\mathbf{q}}_p\|_F^2, \quad p = 1, 2, \dots, P \quad (27)$$

where $\hat{\mathbf{q}}_p = -\text{angle}(\text{normal}(\hat{\mathbf{b}}_t(r_p)))$. Consequently, the estimate of κ_p is obtained as $\kappa_p = [\kappa_{p1}, \kappa_{p2}] = (\mathbf{O} \mathbf{O}^T)^{-1} \mathbf{O} \hat{\mathbf{q}}_p$. Finally, the range estimate is given by:

$$\hat{r}_p = -\frac{\kappa_{p2} c}{4\pi \Delta f} \quad (28)$$

IV. NUMERICAL SIMULATION

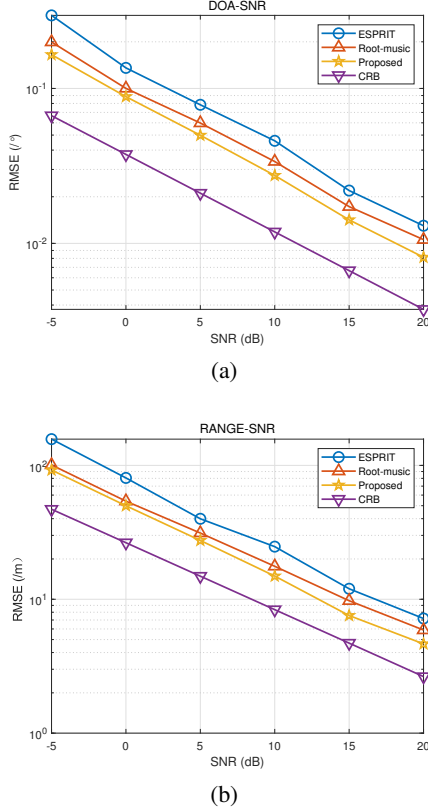


Fig. 2. RMSE under different SNR conditions. (a) RMSE for angle estimation. (b) RMSE for range estimation.

In this section, the proposed method is benchmarked against root-MUSIC [7] and ESPRIT [8]. We employ the root mean square error (RMSE) as the performance metric, formulated as:

$$RMSE_{\theta} = \sqrt{\frac{1}{WP} \sum_{w=1}^W \sum_{p=1}^P \left[\left(\hat{\theta}_{w,p} - \theta_p \right)^2 \right]} \quad (29a)$$

$$RMSE_r = \sqrt{\frac{1}{WP} \sum_{w=1}^W \sum_{p=1}^P \left[\left(\hat{r}_{w,p} - r_p \right)^2 \right]} \quad (29b)$$

where $W = 200$ indicates the number of Monte Carlo trials. The simulation setup includes three distinct targets positioned at $[-20^\circ, 10 \text{ km}]$, $[15^\circ, 12.5 \text{ km}]$, and $[40^\circ, 14 \text{ km}]$. We utilize an FDA-MIMO transceiver architecture equipped with $M = N = 8$ antenna elements. Furthermore, the system operates with a carrier frequency of $f_0 = 10 \text{ GHz}$ and applies a frequency increment of $\Delta f = 3 \text{ kHz}$.

The first experiment evaluates the estimation accuracy of the proposed method and the comparative methods under different signal-to-noise ratio (SNR) conditions. In this experiment, the number of snapshots is 90, and the SNR varies from -5 dB to 20 dB . As shown in Fig. 2, the proposed method achieves the lowest RMSE, which is attributed to its exploitation of the multi-dimensional structural characteristics in the received

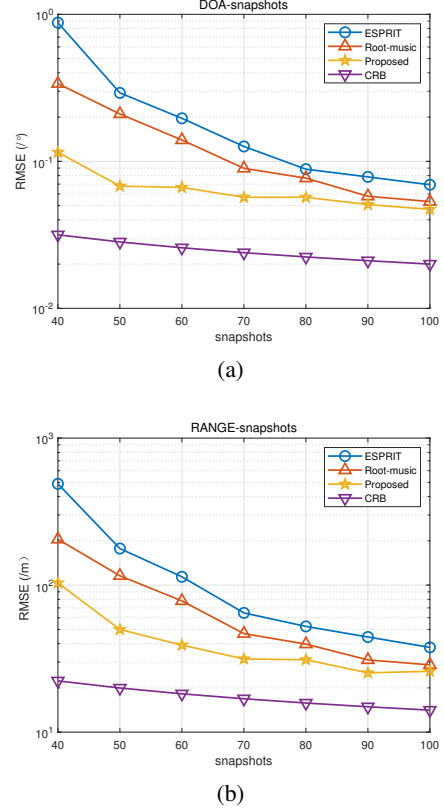


Fig. 3. RMSE under different snapshots conditions. (a) RMSE for angle estimation. (b) RMSE for range estimation.

data. ESPRIT exhibits the poorest performance because it necessitates splitting the array into two overlapping subarrays, thereby reducing the effective array aperture. Furthermore, as the SNR increases, the estimation quality of all algorithms shows evident improvement.

The second experiment evaluates the estimation accuracy of the proposed and comparison methods across different snapshot conditions. In this experiment, the SNR is set to 5 dB , and the number of snapshots varies from 40 to 100. It is evident from Fig. 3 that the RMSE of the proposed method is consistently lower than that of ESPRIT and Root-MUSIC. In addition, the results demonstrate that a larger number of snapshots helps improve the precision of range and angle estimation.

V. CONCLUSION

This paper designs a Root-MUSIC algorithm based on HOSVD that improves the accuracy of angle and range estimation by exploiting the multi-dimensional structural information of the data. The proposed algorithm first obtains the signal subspace through HOSVD. Then, the proposed method leverages the root-MUSIC technique to compute angle estimates in closed form. Finally, the least squares fitting technique is employed to obtain the range estimate. Numerical simulations indicate that the proposed method yields substantially lower parameter estimation errors than existing techniques.

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